AURORA-SR

Connecting RGB Spectral Reconstruction and Physics-Guided Drought Inference with Autonomous Earth-Observation Planning

Technical brief | Dhruv Ranjan | Bengaluru, India | November 2026

STATUS: The perception and mission-planning components have been implemented and evaluated separately, and a software bridge supports forecast-region ranking and replanning. The drought model's confidence output is not yet calibrated as an observation variance. The complete perception-to-planner loop has not been validated end to end or in flight.

1. Research objective

AURORA-SR investigates how uncertainty in Earth-observation measurements could influence a small spacecraft's choice of what to observe next. Rather than ranking targets only by fixed scientific priority, the proposed framework considers expected information gain alongside cloud conditions, energy, onboard storage, execution risk and future access opportunities. The application focus is drought-related environmental monitoring.

2. System architecture

Perception: RGB imagery is mapped to an estimated six-band representation [B, G, R, NIR, SWIR1, SWIR2]. NIR and SWIR are inferred by the model, not directly measured. Derived features include NDVI, NDMI, NDWI, SAVI, EVI and MSI, combined with temporal features in a spatial-temporal drought classifier.
Belief and planning: a Bayesian belief model represents environmental-state uncertainty. Expected information gain is used within a Dynamic Information Utility (DIU) scheduler that considers cloud forecasts, battery, storage and future observation opportunities.
Spacecraft simulation: the research simulator includes two-body/RK4 orbital propagation, Earth rotation and access geometry, eclipse/solar-power estimates, battery and recorder accounting, and ground-station downlink modelling.

3. Component-level results reported in the manuscript

The following evaluations use separate datasets, targets and metrics; they do not constitute one integrated end-to-end score.

Evaluation	Result	Interpretation
RGB-to-six-band reconstruction internal validation	MAE 0.01014 RMSE 0.02566	Pooled normalized reflectance-scale errors.
Drought classifier random split, n=510	ROC-AUC 0.9526 F1 0.9275 at threshold 0.61	Threshold selected on this same partition; labels are mostly proxy-derived and split is not region-disjoint.
Region-held-out run 4 unseen regions, n=56	Learned model: AUC 0.9412, F1 0.6000 Physics baseline: AUC 0.9529, F1 0.7500	Bounded geographic holdout; learned-model F1 is lower than the physics baseline. Small sample.
Synthetic mission simulator nominal, 24 matched runs	DIU final field RMSE 0.01872 95% interval [0.01769, 0.01979]	DIU lowest nominal final RMSE among five policies; myopic policy had slightly higher information gain.

Across four synthetic scenarios, DIU had the lowest mean final RMSE in nominal, battery-stress and cloud-stress cases. Resource-aware myopic scheduling marginally outperformed DIU in storage stress. The paper does not claim universal planner dominance.

4. Principal limitations

• The perception and mission-simulator experiments use different data, targets and metrics; the complete learned perception-to-planner loop has not been evaluated end to end.

• The classifier confidence head is not a calibrated predictive variance and cannot yet be substituted directly into the planner's observation covariance.

• Most cached drought labels are proxy-derived. The principal random split is sample-level rather than geographically disjoint; season-held-out and independent reference-label testing remain necessary.

• The region-held-out result uses only 56 sequences across four unseen regions and shows lower F1 for the learned classifier than for the physics-only baseline.

• The spacecraft simulator is a research model, not flight software. The reported experiment excludes thermal dynamics, attitude slew/dynamics and onboard RGB inference. Higher-fidelity work should address orbital perturbations, attitude/pointing, sensor footprint, battery-temperature behaviour and communications calibration.

5. Next validation and development steps

• Improve label provenance and evaluate against independently referenced drought products or ground observations.

• Run region-disjoint and season-disjoint validation with adequate sample sizes; report per-band reconstruction errors and uncertainty metrics.

• Calibrate predictive uncertainty on a held-out reference set and test reliability, coverage and calibration before using it as planner observation variance.

• Replace the simulator's synthetic measurement process with the learned RGB-to-drought observation model; use matched Monte Carlo seeds and compare full, deterministic and oracle/ablation conditions.

• Increase spacecraft-model fidelity incrementally and validate orbital, pointing, resource and communications models against established references.

6. Feedback sought

I would value expert feedback on: (i) how best to calibrate and propagate spectral/drought uncertainty into an observation covariance; (ii) appropriate validation benchmarks for the spacecraft and planning components; (iii) which spacecraft-physics improvements should be prioritised; and (iv) whether the present scope and evidence are suitable for a public preprint, and what should be strengthened before submission.

7. Project materials

Full manuscript: “AURORA-SR: Connecting RGB Spectral Reconstruction and Physics-Guided Drought Inference with Autonomous Earth-Observation Planning” (32 pages, September 2026).
Code: https://github.com/Dhruv-Ranjan/AURORA_JALADRISHTI
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This brief summarizes the accompanying manuscript. Reported results and limitations are stated as presented in that manuscript; proposed interfaces and future experiments are not represented as completed results.